Formulas for Statistische und Stochastische Grundlagen der Informatik

Universität Stuttgart

Wintersemester 2025/26

Inhaltsverzeichnis

Chapter: 1 Stochastic **4**

1 Finite Probability Space **4**
1.1 Axioms of Probability 4

2 Random Variables **4**

3 Special Concepts **4**
3.1 Laplace Experiment 4
3.2 Indicator Function 4

4 Expected Value **4**
4.1 Properties of Expected Value 5

5 Variance and Standard Deviation **5**
5.1 Properties of Variance 5
5.2 Standard Deviation 5

6 Chebyshev's Inequality **5**
6.1 Intuition 5

7 Independence **5**
7.1 Independence of Events 5
7.2 Independence of Random Variables 5
7.3 Application of independent random variables 6

8 Bayes Theorem **6**

9 Binomial distribution **6**
9.1 Bernoulli Experiment 6

10 Hypergeometrical distribution **6**

11 Discrete Probability Space **6**

12 Geometrical distribution **7**

13 Poisson distribution **7**

14 Law of large numbers **7**
14.1 Weak law of Large numbers 7

15 Central Limit Theorem **8**

16 General Models **8**
16.1 General Probability Space . . . 8
16.1.1 Distribution Function . . . 9
16.1.2 Density Function . . . 9
16.2 Uniform Distribution . . . 9
16.3 Exponential Distribution . . . 10
16.4 Normal Distribution . . . 10
16.4.1 Quantiles . . . 10
16.4.2 $k\sigma$-Ranges . . . 10
16.5 Chi-Square Distribution . . . 11
16.5.1 Student's t-Distribution . . . 11
16.5.2 Fisher's F-Distribution . . . 11

17 Multidimensional random variables **11**
17.1 Marginal distribution (Randverteilung) . . . 11
17.2 Conditional probability function . . . 12
17.3 Total distribution function . . . 12
17.4 Marginal distribution function . . . 12
17.5 Two dimensional continous random variables . . . 12
17.5.1 Combined continous distribution and density function . . . 12
17.5.2 Marginal density . . . 12
17.5.3 Combined Distribution Function . . . 12
17.5.4 Conditional Density . . . 12
17.5.5 Covariance . . . 12
17.5.6 Correlation . . . 12
17.6 2D Normal distribution . . . 13

Chapter: 2 Statistik **14**

18 Basic Concepts (translated) **14**

19 Types of statistic **14**

20 Scales and characteristics **14**
20.1 Characteristic types . . . 14
20.2 Scales of characteristics . . . 14

21 Obtaining Data **14**
21.1 No data yet . . . 14
21.2 Already have data . . . 14
21.2.1 Complete survey vs. sample survey . . . 15

22 Univariate descriptive and explorative statistics **15**
22.1 Initial parameters: frequencies . . . 15
22.2 Graphical methods for univariate data . . . 15
22.3 Characterization of empirical distributions . . . 16
22.3.1 Characterization over Form Parameters . . . 16
22.3.2 Characterization over empirical distribution function . . . 16
22.3.3 Characterization over center of distribution / location parameters . . . 16
22.3.4 Characterization over dispersion measures . . . 17
Quantil (used later): . . . 17
Graphical Representation - expanded box plot . . . 17
Further dispersion measures . . . 17

Transformation rule . . . 18
Shift Theorem . . . 18
22.3.5 Characterization over form parameters . . . 18
Skewness (Schiefe) (moment coefficient) . . . 18
Curvature (Wölbung) (Fisher's curvature measure) . . . 18
22.3.6 Characterization over Konzentrationsmaße . . . 18
Lorenz Curve: . . . 18
Gini-Coefficient . . . 19

23 Multivariate descriptive and explorative statistics 19
23.1 2D Contingency Tables . . . 19
Conditioned relative probability distribution . . . 19
23.2 Context analysis . . . 19
23.2.1 χ^2 coefficient . . . 19
23.2.2 Contingency coefficient . . . 19
23.2.3 Corrected contingency coefficient . . . 20
23.3 Analysis of Quantitative Characteristics . . . 20
23.3.1 Empirical Correlation Coefficient . . . 20
23.4 Linear Regression . . . 20
23.4.1 Method of the smallest squares . . . 20
23.4.2 Residual Analysis . . . 20
Coefficient of determination - Bestimmtheitsmaß . . . 21
23.5 QQ-Plot (Quantil-Quantil Diagram) . . . 21

Chapter 1: Stochastic

1 Finite Probability Space

A finite probability space is defined as a tuple:

$$(\Omega, \mathbb{P}) \quad \text{where} \quad 0 < |\Omega| < \infty$$

with the probability measure:

$$\mathbb{P} : \mathcal{P}(\Omega) \to \mathbb{R}$$

1.1 Axioms of Probability

The probability measure $\mathbb{P}$ satisfies the following properties:

1. **Non-negativity:**
$$\forall A \subseteq \Omega : \mathbb{P}(A) \geq 0$$
2. **Normalization:**
$$\mathbb{P}(\Omega) = 1$$
3. **Additivity for disjoint events:**
$$\forall A, B \subseteq \Omega,\ A \cap B = \emptyset : \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

Note: All other properties of probability can be derived from these axioms.

2 Random Variables

A **random variable** is a function:

$$X : \Omega \to \mathbb{R}$$

that tracks a particular property of the probability space.

A **distribution function** describes the probability of each value of the random variable. The cumulative distribution function follows trivially.

3 Special Concepts

3.1 Laplace Experiment

In a Laplace experiment, all elementary events have equal probability.

3.2 Indicator Function

The indicator function for a set A is defined as:

$$\mathbb{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

4 Expected Value

The **expected value** of a random variable X represents its average value:

$$\mathbb{E}(X) := \sum_{w \in \Omega} X(w)\, \mathbb{P}(\{w\}) = \sum_{x \in X(\Omega)} x \cdot \mathbb{P}(X = x)$$

4.1 Properties of Expected Value

1. **Linearity:**
$$\mathbb{E}\left(\sum_{i=1}^{n} \alpha_i X_i\right) = \sum_{i=1}^{n} \alpha_i \, \mathbb{E}(X_i)$$
2. **Expected value of indicator function:**
$$\mathbb{E}(\mathbb{1}_A) = \mathbb{P}(A)$$

5 Variance and Standard Deviation

The **variance** measures the spread of a random variable's distribution (higher variance indicates greater spread):
$$\mathbb{V}(X) := \mathbb{E}\left((X - \mathbb{E}(X))^2\right) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$$

5.1 Properties of Variance

For constants $a, b \in \mathbb{R}$:
$$\mathbb{V}(aX + b) = a^2 \, \mathbb{V}(X)$$

5.2 Standard Deviation

The **standard deviation** is defined as:
$$\sigma(X) := \sqrt{\mathbb{V}(X)}$$

6 Chebyshev's Inequality

Chebyshev's inequality provides a conservative bound on the probability of deviation from the expected value:
$$\mathbb{P}\left(|X - \mathbb{E}(X)| \geq \epsilon\right) \leq \frac{\mathbb{V}(X)}{\epsilon^2}$$

6.1 Intuition

- Higher variance implies larger possible bounds, as a more spread-out distribution increases the likelihood of values lying outside a given range.
- The bound is inversely proportional to ϵ^2: increasing the tolerance ϵ decreases the probability of exceeding it.

7 Independence

7.1 Independence of Events

Two events A and B are **independent** if:
$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

7.2 Independence of Random Variables

Random variables $X_1, \ldots, X_n$ are **independent** if for all sets $B_1, \ldots, B_n$:
$$\mathbb{P}(X_1 \in B_1, \ldots, X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i)$$

7.3 Application of independent random variables

For independent random variables

$$\mathbb{E}(X \cdot Y) = \mathbb{E}(X)\mathbb{E}(Y)$$

$$\mathbb{V}(\sum_{j=1}^{n} X_j) = \sum_{j=1}^{n} \mathbb{V}(X_j)$$

8 Bayes Theorem

Only applies if all A_i are independent and they form a complete partition of Ω

$$\mathbb{P}(A_j|B) = \frac{\mathbb{P}(B|A_j)\mathbb{P}(A_j)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_j)\mathbb{P}(A_j)}{\sum_{i=1}^{n}\mathbb{P}(B|A_i)\mathbb{P}(A_i)}$$

9 Binomial distribution

9.1 Bernoulli Experiment

A bernoulli experiment is a random variable X where $X(\Omega) = \{0, 1\}$ with $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$

if we chain these bernoulli experiments we get a binomial distribution

$$X \sim Bin(n, p)$$

p probability of success, n amount of repetitions

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mathbb{E}(X) = np$$

$$\mathbb{V}(X) = np(1-p)$$

10 Hypergeometrical distribution

This is based on having r 'red' balls, s 'white' balls, and we pull n balls without returning.

$$X \sim Hyp(n, r, s)$$

The probability of k red balls getting pulled is how many combinations you can get of k red balls within the total amount of balls, times the combinations of n-k white balls with the total amount of white balls. Divided by the total amount of ball combinations

$$\mathbb{P}(X = k) = \frac{\binom{r}{k}\binom{s}{n-k}}{\binom{r+s}{n}}$$

$$\mathbb{E}(X) = n\frac{r}{r+s}$$

11 Discrete Probability Space

Same as the finite, just that the space can be infinite. Same rules apply, the expected value can not converge sometimes

12 Geometrical distribution

If you have a bernoulli experiment, and you want to compute how long it takes until a success, this models it.

$$X \sim G(p)$$
$$\mathbb{P}(X = k) = (1-p)^k p$$
$$\mathbb{E}(X) = \frac{1}{p} - 1$$
$$\mathbb{V}(X) = \frac{1-p}{p^2}$$
$$F^X(i) = 1 - (1-p)^{i+1}$$

This counts how many fails till a success

Note: This distribution has no memory, it doesnt follow the gamblers fallacy. Whatever has already happened doesnt influence the future

13 Poisson distribution

In a set time space, how often does a random event occur? Whats the probability of it happening k times?

$$X \sim Po(\lambda), \lambda > 0$$

Lambda is the average amount of times that happens in a fixed time window

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$
$$\mathbb{E}(X) = \mathbb{V}(X) = \lambda$$

$$X \sim Po(\lambda), Y \sim Po(\mu) \Rightarrow X + Y \sim Po(\lambda + \mu)$$

14 Law of large numbers

14.1 Weak law of Large numbers

Let $X_1, x_2, ..$ be a sequence of independent random variables to the same probability space

All of them have the same expected value μ and the same variance σ

We define the arithmetical mean of the first n random variables as

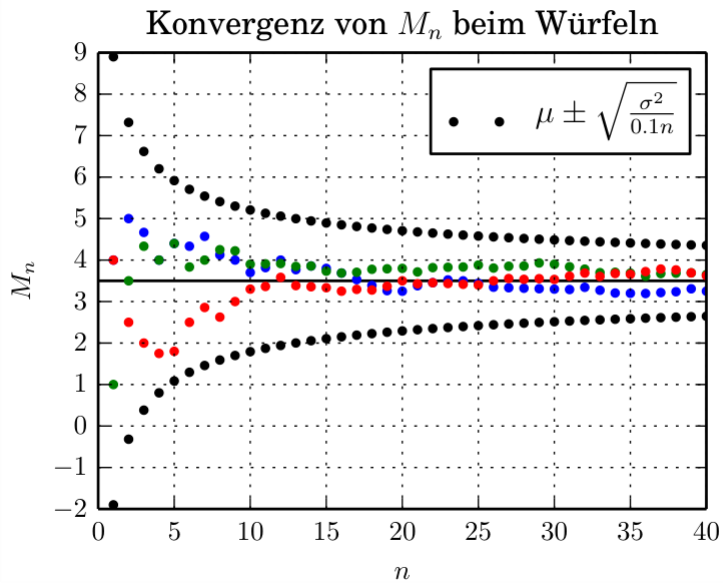
$$M_n := \frac{1}{n} \sum_{j=1}^{n} X_j$$

and $\epsilon > 0$

$$\lim_{n \to \infty} \mathbb{P}(|M_n - \mu| \geq \epsilon) = 0$$

As n grows larger, deviations from the true expected value grow increasingly smaller.

This can also be seen bound with the Chebyshev inequality, As we see, its apretty generous bound



15 Central Limit Theorem

Let $S_n \sim \text{Bin}(n, p)$ with a fixed $0 < p < 1$ and $q := 1 - p$.

We define

$$S_n^* = \frac{S_n - np}{\sqrt{npq}}$$

as the normalized binomial distribution, with expected value 0 and variance 1.

For all $a, b \in \mathbb{R}$ with $a < b$:

$$\lim_{n\to\infty} \mathbb{P}(a \le S_n^* \le b) = \int_a^b \varphi(x)\,dx$$

$$\lim_{n\to\infty} \mathbb{P}(S_n^* \le b) = \int_{-\infty}^b \varphi(x)\,dx$$

where φ is the Gaussian bell function (standard normal density):

$$\varphi(x) := \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Essentially, this states that when the normalized binomial distribution has infinitely large n, it closely resembles a continuous function. Therefore, to compute the cumulative distribution function, integration is sufficient.

16 General Models

16.1 General Probability Space

A probability space is a triple $(\Omega, \mathcal{A}, \mathbb{P})$ where:

- $\Omega \neq \emptyset$ (sample space)
- $\mathcal{A}$ is a σ-algebra on Ω with:

- $\Omega \in \mathcal{A}$
- $A \in \mathcal{A} \Rightarrow \overline{A} := \Omega \setminus A \in \mathcal{A}$

- $\mathbb{P} : \mathcal{A} \to [0, 1]$ is a probability measure with:
  - $\forall A \in \mathcal{A} : \mathbb{P}(A) \geq 0$
  - $\mathbb{P}(\Omega) = 1$

We define a **random variable** X as a function $X : \Omega \to \mathbb{R}$ with the measurable property:

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{A}, \quad \forall x \in \mathbb{R}$$

Informally, this means that for any real number x, the set of all outcomes where the random variable X takes a value less than or equal to x must be an event we can assign a probability to (i.e., it must be in our σ-algebra $\mathcal{A}$).

16.1.1 Distribution Function

$$F : \mathbb{R} \to [0, 1]$$
$$F(x) := \mathbb{P}(X \leq x), \quad \forall x \in \mathbb{R}$$

Properties: F is monotonically increasing and right-continuous, with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$.

16.1.2 Density Function

For continuous probability spaces, the probability at any specific point is zero, so we use a different representation that describes the probability in an **area**.

$$f : \mathbb{R} \to \mathbb{R}_{\geq 0}$$

The total area under the density function equals 1, and it is tightly related to the distribution function:

$$F(x) = \int_{-\infty}^{x} f(t)\, dt$$

Expected value and variance:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$\mathbb{V}(X) = \int_{-\infty}^{\infty} (x - \mathbb{E}(X))^2 \cdot f(x)\, dx = \mathbb{E}\left[(X - \mathbb{E}(X))^2\right] = \sigma^2(X)$$

16.2 Uniform Distribution

A random variable is uniformly distributed, $X \sim \mathcal{U}(a, b)$, if:

$$f(x) := \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}(X) = \frac{a + b}{2}$$

$$\mathbb{V}(X) = \frac{(b - a)^2}{12}$$

16.3 Exponential Distribution

A random variable is exponentially distributed with parameter $\lambda > 0$, denoted $X \sim \text{Exp}(\lambda)$, if:

$$f(x) := \begin{cases} 0 & x \leq 0 \\ \lambda e^{-\lambda x} & x > 0 \end{cases}$$

Distribution function:

$$F(x) := \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x > 0 \end{cases}$$

$$\mathbb{E}(X) = \frac{1}{\lambda}$$

$$\mathbb{V}(X) = \frac{1}{\lambda^2}$$

16.4 Normal Distribution

A random variable is normally distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted $X \sim \mathcal{N}(\mu, \sigma^2)$, if:

$$f(x) := \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

This somewhat resembles the binomial distribution. With parameters $(0, 1)$, we obtain the Gaussian bell function.

We call $Z \sim \mathcal{N}(0, 1)$ the **standard normal distribution**.

Also, we can transform: $X := \sigma \cdot Z + \mu$.

$$\mathbb{E}(X) = \mu$$

$$\mathbb{V}(X) = \sigma^2$$

16.4.1 Quantiles

Quantiles represent the x-coordinate that corresponds to a given percentage of the cumulative distribution function:

$$x_p = (F^X)^{-1}(p) = \sigma z_p + \mu$$

where z_p is the p-quantile of the standard normal distribution.

Important p-quantiles for standard normal distribution:

p	50%	75%	90%	95%	97.5%	99%
z_p	0.00	0.67	1.28	1.64	1.96	2.33

16.4.2 $k\sigma$-Ranges

The inverse of quantiles: you specify a number k corresponding to the width of the interval (in multiples of σ) in each direction from the mean, and you get the probability:

$$\mathbb{P}(\mu - k\sigma \leq X \leq \mu + k\sigma)$$

16.5 Chi-Square Distribution

Let $X_i, i = 1, \ldots, n$ be independent and identically distributed random variables following the standard normal distribution.

Then we call $Z = X_1^2 + \cdots + X_n^2$ **chi-square distributed** with n degrees of freedom:

$$Z \sim \chi^2(n)$$
$$\mathbb{E}(Z) = n$$
$$\mathbb{V}(Z) = 2n$$

16.5.1 Student's t-Distribution

Let X be standard normally distributed and $Z \sim \chi^2(n)$ be independent. Then

$$T = \frac{X}{\sqrt{Z/n}}$$

is called **t-distributed** with n degrees of freedom:

$$T \sim t(n)$$

$$\mathbb{E}(T) = 0, \quad (n \geq 2)$$
$$\mathbb{V}(T) = \frac{n}{n-2}, \quad (n \geq 3)$$

16.5.2 Fisher's F-Distribution

Let $X \sim \chi^2(m)$ and $Y \sim \chi^2(n)$ be independent. Then

$$Z = \frac{X/m}{Y/n}$$

is **Fisher distributed** with m and n degrees of freedom:

$$Z \sim F(m, n)$$

$$\mathbb{E}(Z) = \frac{n}{n-2}, \quad n \geq 3$$
$$\mathbb{V}(Z) = \frac{2n^2(m+n-2)}{m(n-4)(n-2)^2}, \quad n \geq 5$$

17 Multidimensional random variables

$$\mathbb{P}^{(X,Y)}(\{(x,y)\}) = f(x,y) := \mathbb{P}(X = x, Y = y)$$

Note: $f(x,y)$ doesnt mean the function is continous, its just the discrete equivalent

17.1 Marginal distribution (Randverteilung)

$$f_X(x) = \mathbb{P}(X = x) = \sum_{j=1,2,..} f(x, y_j)$$

Analog für Y

17.2 Conditional probability function

$$f_X(x|y) = \frac{f(x,y)}{f_Y(y)}$$

17.3 Total distribution function

$$F(x,y) = \mathbb{P}(X \leq x, Y \leq y) = \sum_{x_i \leq x} \sum_{y_j \leq y} f(x_i, y_j)$$

17.4 Marginal distribution function

$$F_X(x) = \mathbb{P}(X \leq x) = F(x, \infty) = \sum_{x_i \leq x} \sum_{j=1,2,\ldots} f(x_i, y_j)$$

17.5 Two dimensional continous random variables

17.5.1 Combined continous distribution and density function

We call X,Y continous if there exists a two dimensional density function $f(x,y) \geq 0$ with

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x,y) dy dx$$

17.5.2 Marginal density

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy$$

Total area at a point in x

17.5.3 Combined Distribution Function

$$F(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v) dv du$$

17.5.4 Conditional Density

$$f_X(x|y) = \frac{f(x,y)}{f_Y(y)}$$

17.5.5 Covariance

It represents the degree to which two random variables vary together. A positive covariance indicates that the variables tend to move in the same direction, while a negative covariance indicates they tend to move in opposite directions. A covariance of zero suggests no linear relationship.

$$C(X,Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y)))$$

17.5.6 Correlation

It represents the standardized measure of linear association between two random variables, normalized to always lie between -1 and 1. A correlation of +1 indicates perfect positive linear relationship, -1 indicates perfect negative linear relationship, and 0 indicates no linear relationship. Unlike covariance, correlation is dimensionless and scale-independent.

$$r = r(X,Y) = \frac{C(X,Y)}{\sqrt{\mathbb{V}(X)\mathbb{V}(Y)}} = \frac{C(X,Y)}{\sigma_X \sigma_Y}$$

17.6 2D Normal distribution

Complex formula hopefully it doesnt come in the exam

Chapter 2: Statistik

18 Basic Concepts (translated)

- Statistische Einheit - Object that we analyze
- Population - Set of all those objects
- Stichprobe - Subset of the population we are actually analizing
- Merkmal - Characteristic we are checking
- Wert - Value of the characteristic for the object
- Kenngröße - Parameters
- Lagemaße - Dimensions

19 Types of statistic

- Descriptive Statistic: Condensing information, no stochastic needed
- Explorative Statistic: Looking for structures and such in the data
- Inductive Statistic: Reach conclusions based on the data, uses stochastic and such

20 Scales and characteristics

20.1 Characteristic types

- Discrete: Countable - Amount of workers in a company
- Continous: Values from an interval - Height of the workers
- Semi-Continous: Continous, but not continously measured - Age of workers

20.2 Scales of characteristics

- Nominal Scale : Names - Categories
- Ordinal Scale: Ordered, but distance cant be measured
- Cardinal Scale: Ordered and distance can be measured

21 Obtaining Data

21.1 No data yet

Conduct an experiment

21.2 Already have data

Survey: collection of already existing data

1. Survey for the data: primary statistics
2. Survey based on existing (raw) data: secondary statistics
3. Survey based on existing compressed/transformed data: tertiary statistics

21.2.1 Complete survey vs. sample survey

One is harder to do, the other can cause bias/incorrect output data

22 Univariate descriptive and explorative statistics

Explorative and descriptive statistics are sometimes hard to distinguish, so they are treated together

Example: Measuring Clients per day

Objective: Describe the data, compress it, classify it and such

22.1 Initial parameters: frequencies

We have a feature X observed in n Objects

Raw data

$$x_1, ..., x_n$$

Classify by frequency Observed values

$$X : a_1, .., a_k | k \leq n$$

We define

- $h(a_j) = h_j$ Absolute frequency
- $f(a_j) = f_j = \frac{h_j}{n}$ Relative frequency
- $h_1, .., h_k$ Absolute frequency distribution
- $f_1, .., f_k$ Relative frequency distribution

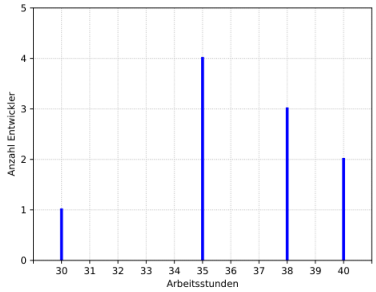
usually represented in a histogram

Usual rule of thumb

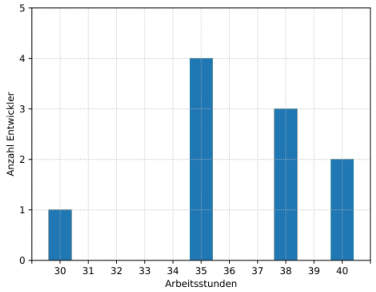
$$k = \lfloor \sqrt{n} \rfloor \text{ for } n \leq 100$$
$$k = \lfloor 10 \log_{10}(n) \rfloor$$

22.2 Graphical methods for univariate data

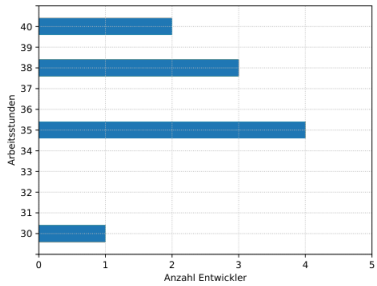
- Strichliste - Tally chart
- Stabdiagramm - Bar chart
- Säulen/-Balkendiagramm - Column/bar chart
- Histogramm - Histogram
- Kreisdiagramm - Pie chart



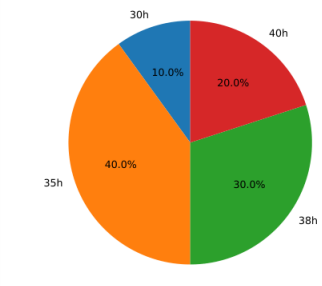
(a) Stabdiagramm



(b) Säulendiagramm



(c) Balkendiagramm



(d) Kreisdiagramm

22.3 Characterization of empirical distributions

22.3.1 Characterization over Form Parameters

- Amount of peaks: uni-,bi-,multimodal
- Steigung: linkssteil/rechtsschief

22.3.2 Characterization over empirical distribution function

ONLY FOR ORDINAL SCALED DATA

$$H(x) = \sum_{i:a_i \leq x} h_j \text{ : absolute cumulative frequency distribution}$$

$$F(x) = \sum_{i:a_i \leq x} f_j \text{ : relative cumulative frequency distribution}$$

(empirical distribution function)

22.3.3 Characterization over center of distribution / location parameters

Lagemaße := $L : \mathbb{R}^n \to \mathbb{R}$

$$L(x_1 + a, ..., x_n + a) = L(x_1, .., x_n) + a : \forall a, x_1, .., x_n \in \mathbb{R}$$

Describes the center of a distribution through a numerical value
Examples of center of the distribution

- arithmetic mean $\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$
- median x_{med} = the one at the middle or the mean between the two in the middle

- mode: most freqeuent elemetn
- Geometrical mean: $x_{geom} = (x_1 \cdot ... \cdot x_n)^{(\frac{1}{n})}$ç
- Harmonical mean: $x_{harm} = \frac{1}{\frac{1}{n}\sum_{i=1}^{n}\frac{1}{x_j}}$

They can also be described as optimality equations

- $\overline{x}$ minimizes $Q(z, x_1, ..., x_n) = \sum_{i=1}^{n}(x_i - z)^2$
- x_{med} minimizes $Q(z, x_1, ..., x_n) = \sum_{i=1}^{n}|x_i - z|$
- x_{mod} minimizes $Q(z, x_1, ..., x_n) = \sum_{i=1}^{n} I(x_i, z)$
  With $I(x_i, z)$ being the indicator function which equals 1 if $x_i \neq z$ and 0 otherwise

22.3.4 Characterization over dispersion measures

Streuengsmaß:= $S : \mathbb{R}^n \to \mathbb{R}$

$$S(x_1 + a, .., x_n + a) = S(x_1, .., x_n) \forall a, x_1, .., x_n \in \mathbb{R}$$

So somewhat like a number that represents the dispersion of the data, for example shifting the data by a doesnt affect it, same dispersion

Quantil (used later): its like the inverse of the distribution function

$$x_p = F^{-1}(p)$$

Basically, its a **hypothetical** value that splits the data (ordered) in two, the left side has values $\leq x_p$ with at least p of the data and the right side has values $\geq x_p$with at least $1 - p$
NOT DISJOINT
Dispersion measures

- Interquartilabstand (IQR): $d_Q = x_{0.75} - x_{0.25}$

We can have a 5-point summary to describe its dispersion

$$x_{min}, x_{0.25}, x_{med}, x_{0.75}, x_{max}$$

Graphical Representation - expanded box plot normal box plot is just the expanded version without the zaun
We draw a rectangle with right edge in $x_{0.75}$, left edge in $x_{0.25}$ and a like in x_{med}
Then we add two vertical lines in the values closest to the center such as

$$[z_u, z_o] = [x_{0.25} - 1.5d_Q, x_{0.75} + 1.5d_Q]$$

There shouldnt be to many points outside of fence, if any

Further dispersion measures

- Empirical Variance: $\tilde{s}^2 = \frac{1}{n}\sum(x_i - \overline{x})^2$
- Stichprobevarianz: $s^2 = \frac{1}{n-1}\sum(x_i - \overline{x})^2$
- Variationskoeffizient: $\tilde{v} = \frac{\tilde{s}}{\overline{x}}$
- Empirische standardabweichung: $\tilde{s} = \sqrt{\tilde{s}^2}$

The -1 in the Stichprobevarianz comes from that we already know the mean, so we lose one degreee of freedom, so that corrects it.

Transformation rule : For a linear transformation $y = ax + b$

$$\tilde{s}_y^2 = a^2 \tilde{s}_x^2 \text{ bzw } \tilde{s}_y = |a| \tilde{s}_x$$

This translates to that if data is multiplied by a and shifted by b, the effect on the variance of the shifting dissapears in the dispersion, and the multiplication is squared

The effect on the standardabweichung of shifting is also null and on the multiplication is scaled by $|a|$

Shift Theorem : $\forall c \in \mathbb{R}$

$$\sum_{i=1}^{n} (x_i - c)^2 = \sum_{i=1}^{n} (x_i - \overline{x})^2 + n(\overline{x} - c)^2$$

If we subtitute for c = 0 and divide by n

$$\tilde{s}^2 = \left(\frac{1}{n} \sum_{i=1}^{n} x_i^2 \right) - \overline{x}^2$$

This gives us a faster formula for empirical variance, and just generally divides the concept of

The total sum of squared distances to a point c can be divided into the spread around the arithmetical mean and the distance between the mean and c (n times)

22.3.5 Characterization over form parameters

Skewness (Schiefe) (moment coefficient)

$$g_m = \frac{\frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x})^3}{\tilde{s}^3}$$

represents the symmetry of the data

- 0 for symmetry, anything else deviates

Curvature (Wölbung) (Fisher's curvature measure)

$$\gamma = \frac{\frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x})^4}{\tilde{s}^4} - 3$$

Describes how strong the occupation of the central and peripherical areas of the data are

- 0: resembles standard normal distribution
- ¿ 0: spikier, extremes are populated
- ¡ 0: flatter

22.3.6 Characterization over Konzentrationsmaße

Lorenz Curve: Only for cardinalized , non negative characteristics. its a way to visualize concentration/ inequality

$$x_1 \leq ... \leq x_n$$

The curve is the line that goes through the points $(0,0), (u_1, v_1), .., (u_n, v_n)$

With

$$u_j = \frac{j}{n}$$

$$v_j = \frac{\sum_{i=1}^{j} x_i}{\sum_{i=1}^{n} x_i}$$

Basically represents for a money and people statistic: Point (u,v) represents that u percentage of people own v percent of the money.

Always below the diagonal, pure diagonal means complete equality.

Gini-Coefficient

$$G = \frac{\text{surface between diagonal and lorenz curve}}{\text{surface between diagonal and u-axis}} =$$
$$= 2 \cdot \text{ surface between diagonal and lorenz curve} =$$
$$= \frac{2\sum_{i=1}^{n} i x_i}{n\sum_{i=1}^{n} x_i} - \frac{n+1}{n}$$

We can also normalize it

$$G^* = \frac{G}{G_{max}} = \frac{n}{n-1} G \in [0,1]$$

Higher gini Coefficient means more unequal, if zero completely equal

23 Multivariate descriptive and explorative statistics

We are now looking at multiple characteristics per object

23.1 2D Contingency Tables

Way of representing the statistics for 2 variables

	b_1	$\dots$	b_m	
a_1	h_{11}	$\cdots$	h_{1m}	$h_{1\cdot}$
a_2	h_{21}	$\cdots$	h_{2m}	$h_{2\cdot}$
$\vdots$	$\vdots$		$\vdots$	$\vdots$
a_k	h_{k1}	$\cdots$	h_{km}	$h_{k\cdot}$
	$h_{\cdot 1}$	$\dots$	$h_{\cdot m}$	n

Can also be done with relative frequencies instead

Conditioned relative probability distribution

$$f_X(a_i|b_j) = \frac{h_{ij}}{h_{\cdot j}}$$

23.2 Context analysis

23.2.1 χ^2 coefficient

Its defined through

$$\chi^2 = \sum_{i=1}^{k}\sum_{j=1}^{m} \frac{(h_{ij} - \tilde{h}_{ij})^2}{\tilde{h_{ij}}} = \sum_{i=1}^{k}\sum_{j=1}^{m} \frac{(h_{ij} - \frac{h_{i\cdot}h_{\cdot j}}{n})^2}{\frac{h_{i\cdot}h_{\cdot j}}{n}} \in [0,\infty)$$

This represents a discrepancy between X and Y.

- big value : X,Y are dependent on eachother
- small value : are independent

23.2.2 Contingency coefficient

$$K = \sqrt{\frac{\chi^2}{n+\chi^2}} \in \left[0, \sqrt{\frac{M-1}{M}}\right] \text{ with } M = \min\{k,m\}$$

23.2.3 Corrected contingency coefficient

$$K^* = \frac{K}{\sqrt{\frac{M-1}{M}}} \in [0,1]$$

Also represents how X and Y are related

23.3 Analysis of Quantitative Characteristics

23.3.1 Empirical Correlation Coefficient

For a sample size $(x_1, y_1), ..., (x_n, y_n)$

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}} = \begin{cases} r > 0 & \text{positive correlation} \\ r < 0 & \text{negative correlation} \\ r = 0 & \text{no correlation} \end{cases}$$

23.4 Linear Regression

The idea is to model a set of data and 2 characteristics to a linear function $Y = f(X) + \epsilon$
with

$$f(X) = \alpha + \beta X$$

$$y_i = \alpha + \beta x_i + \epsilon_i$$

To get the most accurate function, we want to minimize each ϵ_i, for that we have different methods

23.4.1 Method of the smallest squares

$$Q(\alpha, \beta) = \frac{1}{n}\sum_{i=1}^{n}\epsilon_i^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\alpha + \beta x_i))^2$$

We want to minimize Q, so we derive and equal to zero
after deriving (which we can do by hand)

$$\hat{\alpha} = \overline{y} - \hat{\beta}\overline{x}$$

$$\hat{\beta} = \frac{\tilde{s}_{x,y}}{\tilde{s}_x^2}$$

23.4.2 Residual Analysis

Measures how good a regression model is
We define $\hat{y}_i$ has the predicted value by the model

- SQT (Sum of squares total): $\sum(y_i - \overline{y}_i)^2$, how much the data varies around the
- SQE (Sum of squares explained): $\sum(\hat{y}_i - \overline{y}_i)^2$, how much the regression model explains the variation
- SQR (Sum of squares residual): $\sum(y_i - \hat{y}_i)^2$ measures the errors in the variation

$$SQT = SQE + SQR$$

$$\text{Total variability} = \text{explained variability} + \text{unexplained variability}$$

And we can look at how good the regression model is

Coefficient of determination - Bestimmtheitsmaß

$$R^2 = \frac{SQE}{SQT}$$

What percentage of the variation is predicted

- $R^2 = 1$ Perfect fit with the model
- $R^2 = 0$ No relationship, regression model sucks

$R^2 = 0.8$ means 80% of the variability in y is explained by x

$$R^2 = r_{xy}^2$$

r being the empirical correlation coefficient

23.5 QQ-Plot (Quantil-Quantil Diagram)

They are a diagramm where you plot the points (x_p, y_p), representing the p-quantils of two characteristics

Then we can compare it with the analytical distribution

24 Schätzprobleme (Estimating Problems)